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compact1, compact2, compact3

java.util

# Class Spliterators.AbstractSpliterator<T>

java.lang.Object
  java.util.Spliterators.AbstractSpliterator<T>

**All Implemented Interfaces:**

Spliterator<T>

**Enclosing class:**

Spliterators

---

```
public abstract static class Spliterators.AbstractSpliterator<T>
extends Object
implements Spliterator<T>
```

An abstract `Spliterator` that implements `trySplit` to permit limited parallelism.

An extending class need only implement `tryAdvance`. The extending class should override `forEach` if it can provide a more performant implementation.

**API Note:**

This class is a useful aid for creating a spliterator when it is not possible or difficult to efficiently partition elements in a manner allowing balanced parallel computation.

An alternative to using this class, that also permits limited parallelism, is to create a spliterator from an iterator (see `Spliterators.spliterator(Iterator, long, int)`. Depending on the circumstances using an iterator may be easier or more convenient than extending this class, such as when there is already an iterator available to use.

**Since:**

1.8

**See Also:**

Spliterators.spliterator(Iterator, long, int)

## Nested Class Summary

### Nested classes/interfaces inherited from interface java.util.Spliterator

Spliterator.OfDouble, Spliterator.OfInt, Spliterator.OfLong, Spliterator.OfPrimitive<T,T_CONS,T_SPLITR extends Spliterator.OfPrimitive<T,T_CONS,T_SPLITR>>

## Field Summary

### Fields inherited from interface java.util.Spliterator

CONCURRENT, DISTINCT, IMMUTABLE, NONNULL, ORDERED, SIZED, SORTED, SUBSIZED

## Constructor Summary

**Constructors**

| Modifier | Constructor | Description |
| --- | --- | --- |
| protected | **AbstractSpliterator**(long est, int additionalCharacteristics) | Creates a spliterator reporting the given estimated size and additionalCharacteristics. |

## Method Summary

**Concrete Methods**

| Modifier and Type | Method | Description |
| --- | --- | --- |
| int | **characteristics**() | Returns a set of characteristics of this Spliterator an |
| long | **estimateSize**() | Returns an estimate of the number of elements that encountered by a **Spliterator.forEachRemaining(java.util.functi super T>)** traversal, or returns **Long.MAX_VALUE** if in unknown, or too expensive to compute. |

| | | |
|---|---|---|
| Spliterator<T> | trySplit() | If this spliterator can be partitioned, returns a Splite<br>elements, that will, upon return from this method, no<br>this Spliterator. |

### Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

### Methods inherited from interface java.util.Spliterator

forEachRemaining, getComparator, getExactSizeIfKnown, hasCharacteristics, tryAdvance

## *Constructor Detail*

### AbstractSpliterator

```
protected AbstractSpliterator(long est,
                              int additionalCharacteristics)
```

Creates a spliterator reporting the given estimated size and additionalCharacteristics.

**Parameters:**

`est - the estimated size of this spliterator if known, otherwise Long.MAX_VALUE.`

`additionalCharacteristics - properties of this spliterator's source or elements. If SIZED is reported then this spliterator will additionally report SUBSIZED.`

## *Method Detail*

### trySplit

```
public Spliterator<T> trySplit()
```

If this spliterator can be partitioned, returns a Spliterator covering elements, that will, upon return from this method, not be covered by this Spliterator.

If this Spliterator is `Spliterator.ORDERED`, the returned Spliterator must cover a strict prefix of the elements.

Unless this Spliterator covers an infinite number of elements, repeated calls to `trySplit()` must eventually return `null`. Upon non-null return:

- the value reported for `estimateSize()` before splitting, must, after splitting, be greater than or equal to `estimateSize()` for this and the returned Spliterator; and
- if this Spliterator is `SUBSIZED`, then `estimateSize()` for this spliterator before splitting must be equal to the sum of `estimateSize()` for this and the returned Spliterator after splitting.

This method may return `null` for any reason, including emptiness, inability to split after traversal has commenced, data structure constraints, and efficiency considerations. This implementation permits limited parallelism.

**Specified by:**

`trySplit` in interface `Spliterator<T>`

**Returns:**

a Spliterator covering some portion of the elements, or null if this spliterator cannot be split

### estimateSize

```
public long estimateSize()
```

Returns an estimate of the number of elements that would be encountered by a `Spliterator.forEachRemaining(java.util.function.Consumer<? super T>)` traversal, or returns `Long.MAX_VALUE` if infinite, unknown, or too expensive to compute.

If this Spliterator is `Spliterator.SIZED` and has not yet been partially traversed or split, or this Spliterator is `Spliterator.SUBSIZED` and has not yet been partially traversed, this estimate must be an accurate count of elements that would be encountered by a complete traversal. Otherwise, this estimate may be arbitrarily inaccurate, but must decrease as specified across invocations of `Spliterator.trySplit()`.

**Specified by:**

`estimateSize` in interface `Spliterator<T>`

**Implementation Requirements:**

This implementation returns the estimated size as reported when created and, if the estimate size is known, decreases in size when split.

**Returns:**

the estimated size, or Long.MAX_VALUE if infinite, unknown, or too expensive to compute.

### characteristics

```
public int characteristics()
```

Returns a set of characteristics of this Spliterator and its elements. The result is represented as ORed values from `Spliterator.ORDERED`, `Spliterator.DISTINCT`, `Spliterator.SORTED`, `Spliterator.SIZED`, `Spliterator.NONNULL`, `Spliterator.IMMUTABLE`, `Spliterator.CONCURRENT`, `Spliterator.SUBSIZED`. Repeated calls to `characteristics()` on a given spliterator, prior to or in-between calls to `trySplit`, should always return the same result.

If a Spliterator reports an inconsistent set of characteristics (either those returned from a single invocation or across multiple invocations), no guarantees can be made about any computation using this Spliterator.

**Specified by:**

`characteristics` in interface `Spliterator<T>`

**Implementation Requirements:**

This implementation returns the characteristics as reported when created.

**Returns:**

a representation of characteristics



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